

SE Lab 4 — VibeCoding: Chat History Summary

Project: Lunar Lander Repair Lab (Pygame)

This document summarizes the implementation steps discussed in ChatGPT. It is not a verbatim transcript of the full conversation.

Task 1 — Touchdown validation

Updated the successful-landing condition to check that the ship is over a pad and that horizontal speed, vertical speed, and landing angle are all within the constants defined in the starter code.

Task 2 — Fuel-dependent ship colour

Implemented `ship_color(fuel_ratio)` to clamp the fuel ratio between 0 and 1 and interpolate the hull colour so it becomes redder as fuel runs low.

Task 3 — Landing feedback

Implemented `on_landing(score)` to print a distinct message for higher-scoring landings and a safe-landing message otherwise.

Task 4 — Bonus life threshold

Configured `bonus_life_threshold()` to return 1500 points, using the existing bonus-life bookkeeping in the game.

Verification and submission

The provided final source passed Python compilation (`python -m py_compile equivalent`). The user reported that all four tasks were completed and supplied BEFORE and AFTER gameplay recordings. This package uses the first supplied video as BEFORE and the latest supplied video as AFTER.

ChatGPT conversation link

<https://chatgpt.com/c/6ac8fd41-a444-83ee-a390-353225ff12fc>

Important: The lab handout asks for chat history exported as a doc/PDF. This file is a summary only. If the instructor requires the complete verbatim history, export the conversation directly from ChatGPT and add that export.